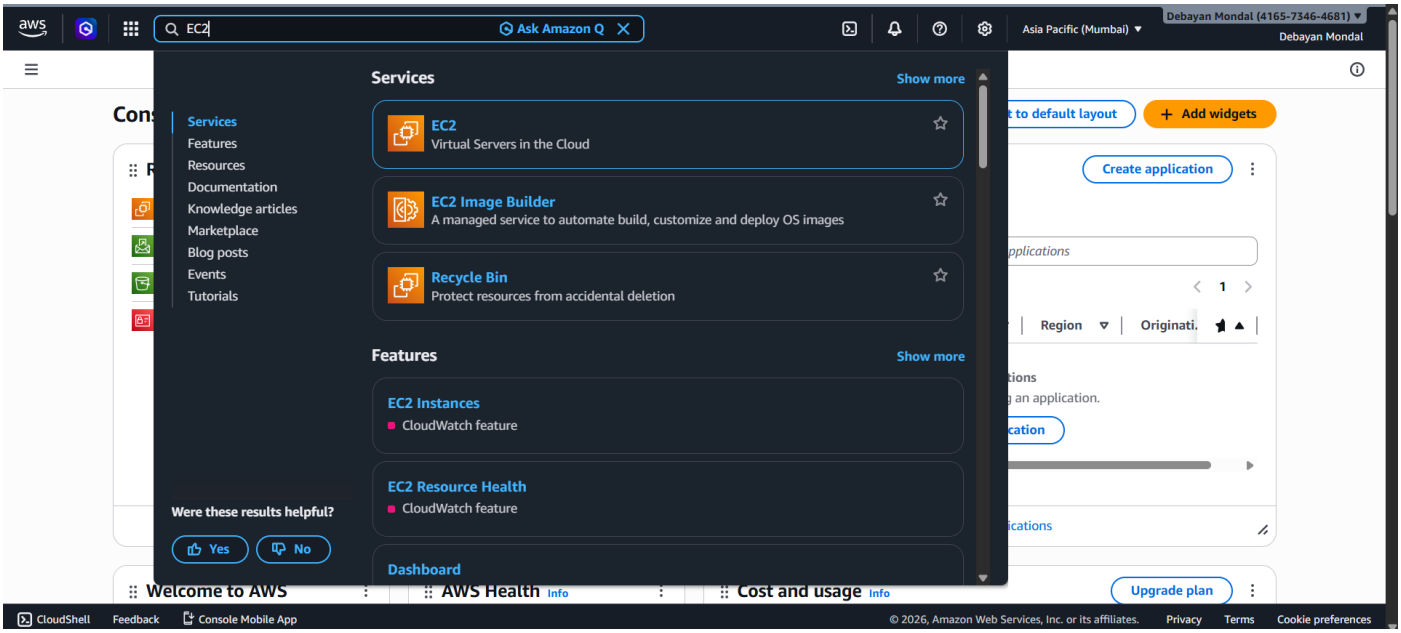
Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Assignment No.:11

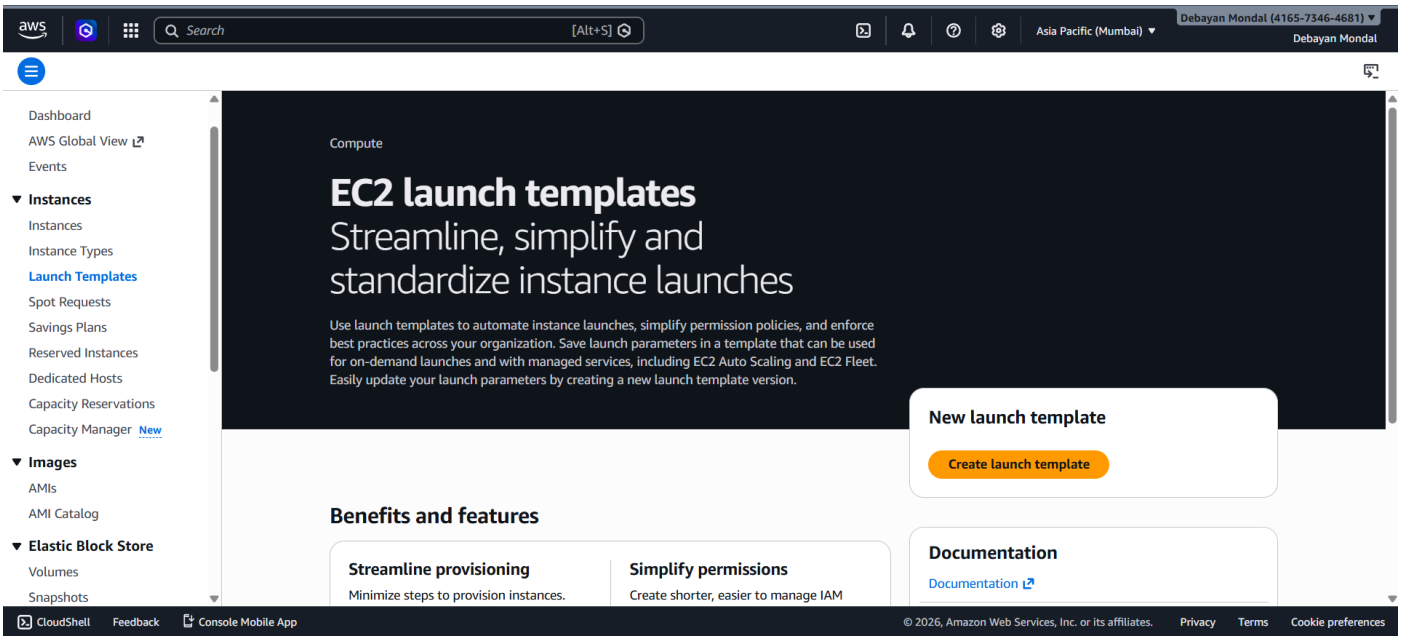
Problem Statement: Build scaling plans in AWS that balance the load on different EC2 instances.

Solution:

Step 1: Search EC2 in the search bar and click on it.



Step 2: Click on **Launch Templates** and click on **Create launch template** button.



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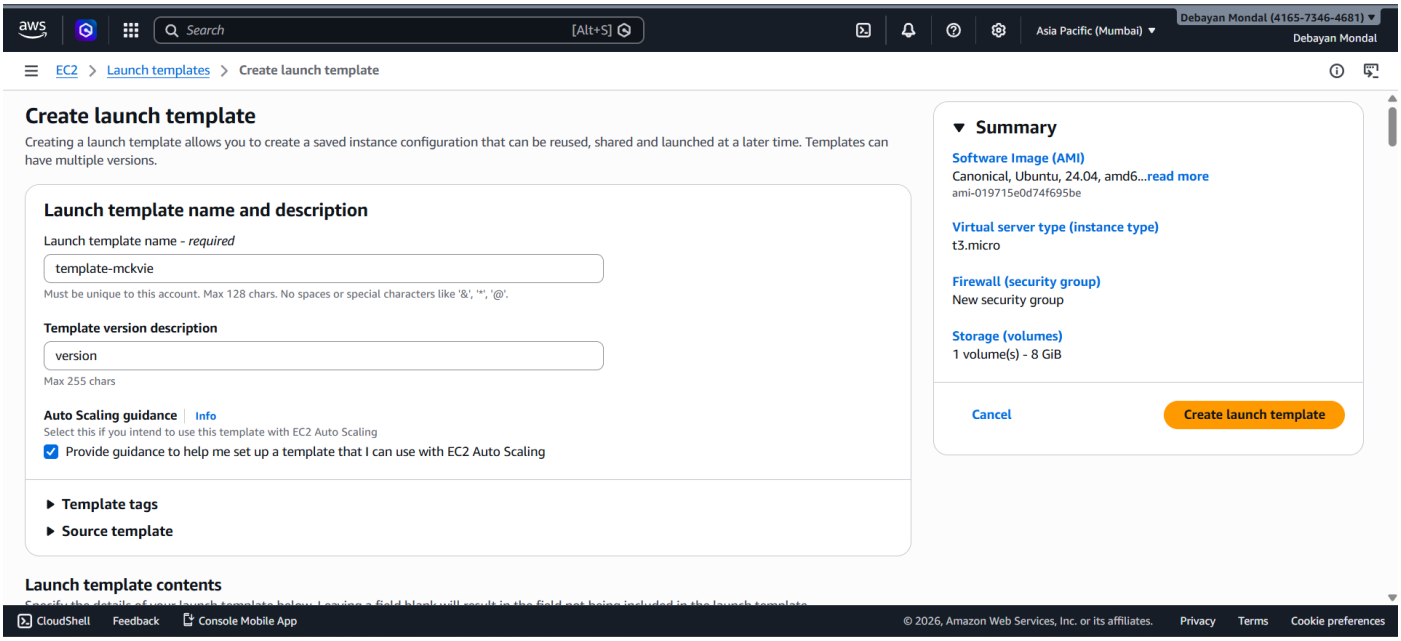
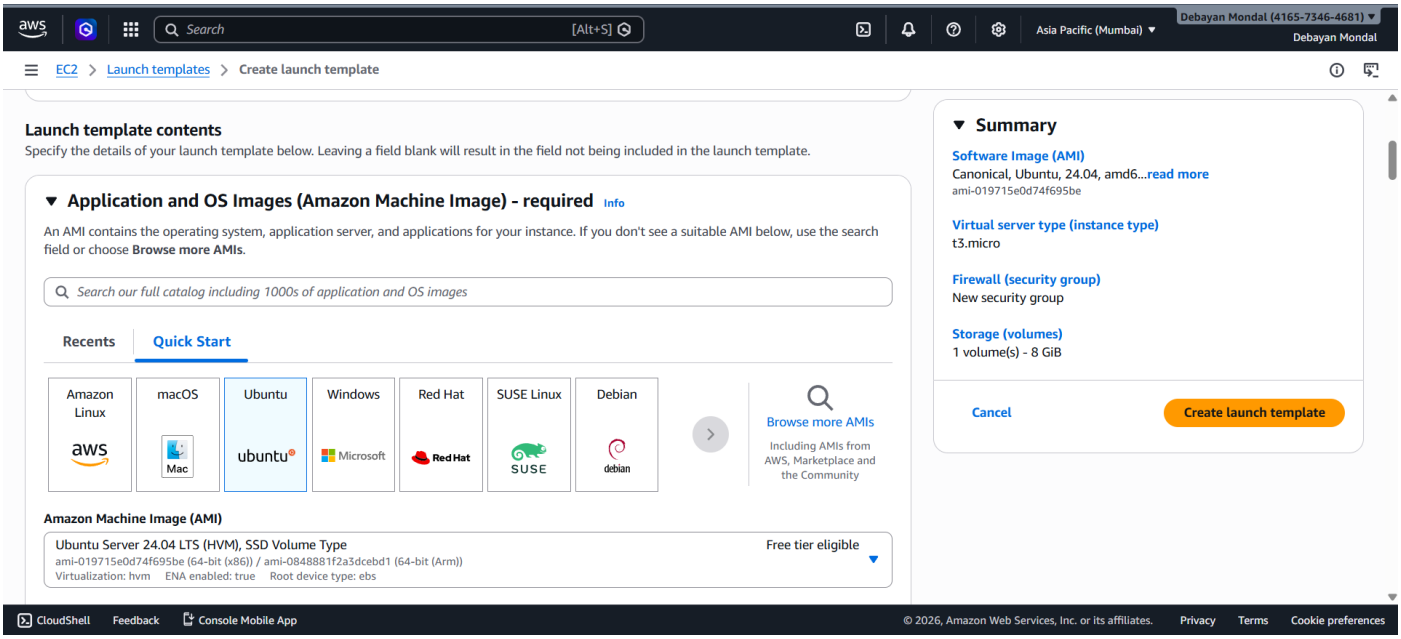
Step 3: Enter the template name and version description and click on the check box **Provide guidance to help me set up a template that I can use with EC2 Auto Scaling**.



Step 4: Click on **Quick Start** and select **Ubuntu**.



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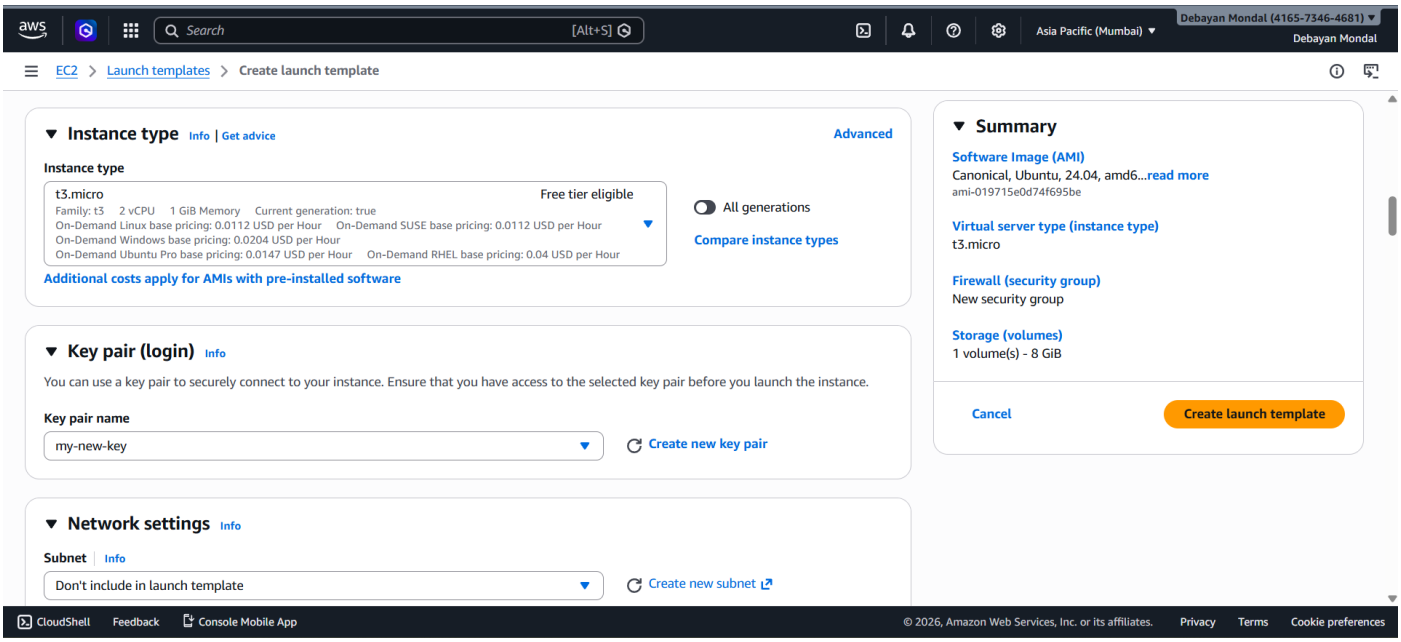
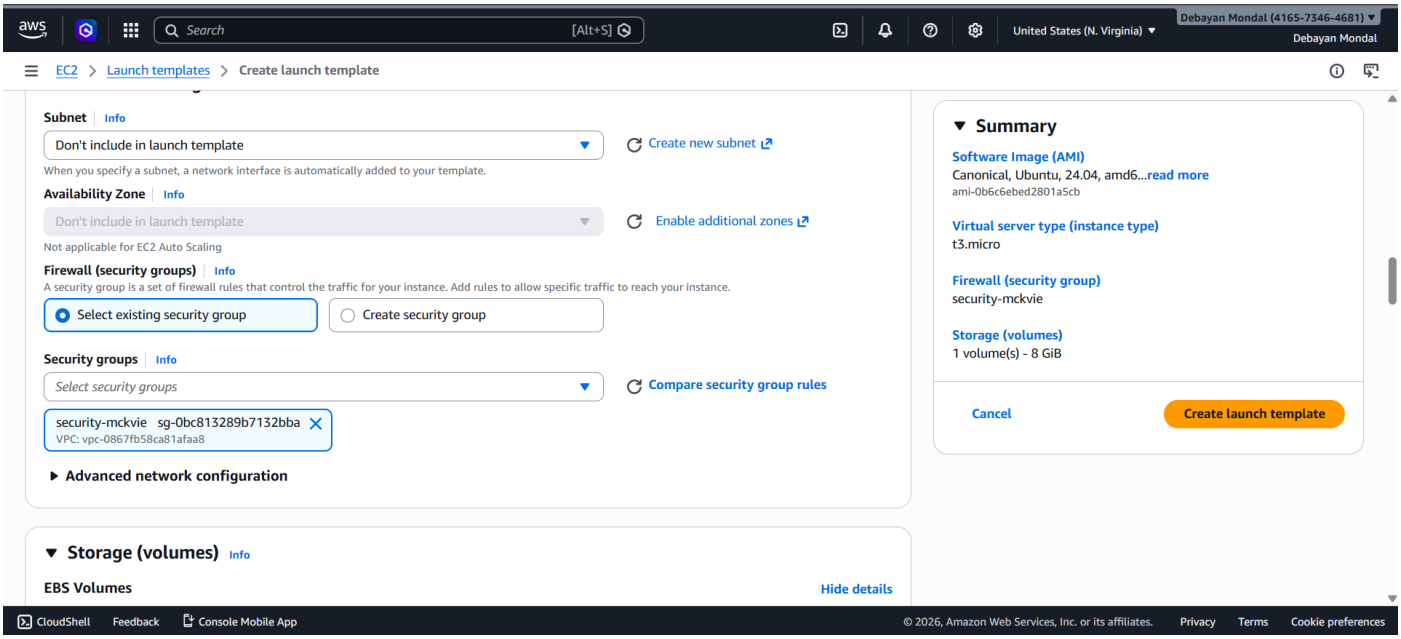
Step 5: From the drop down menu of **Instance type** select **t3.micro** and select an key pair.



Step 6: Click on **Select existing security group** and choose an already existing security group from the drop down.



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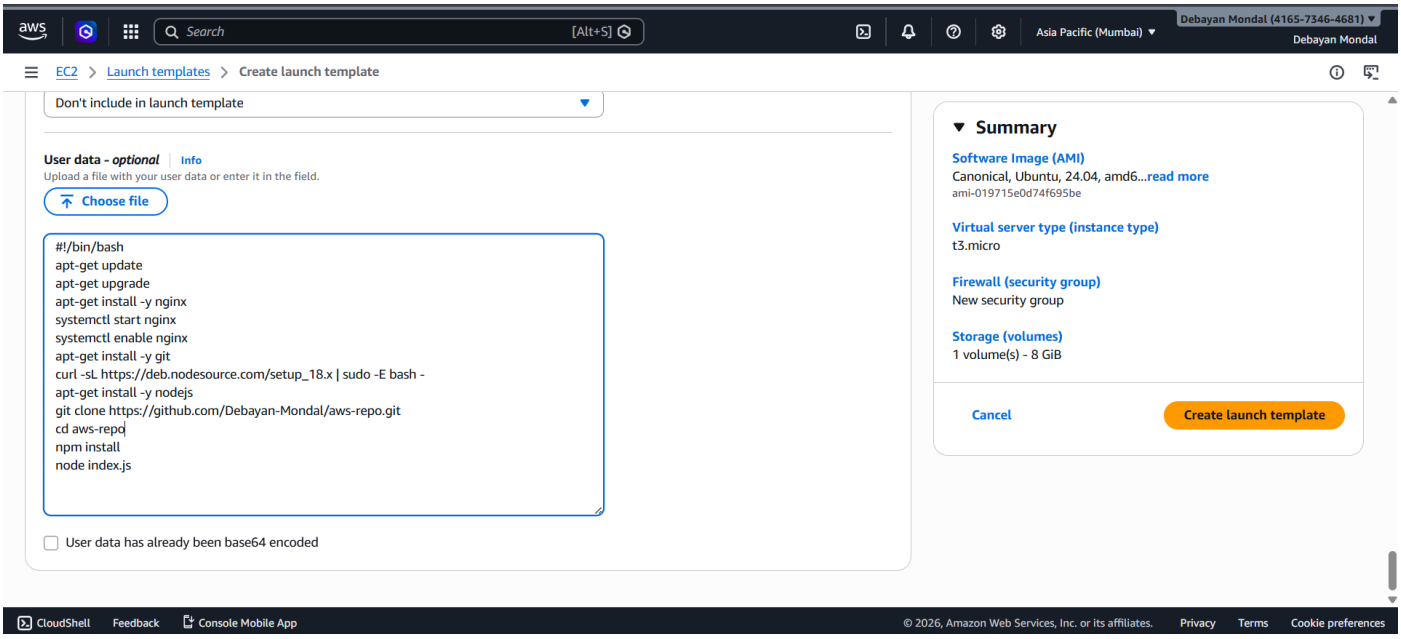
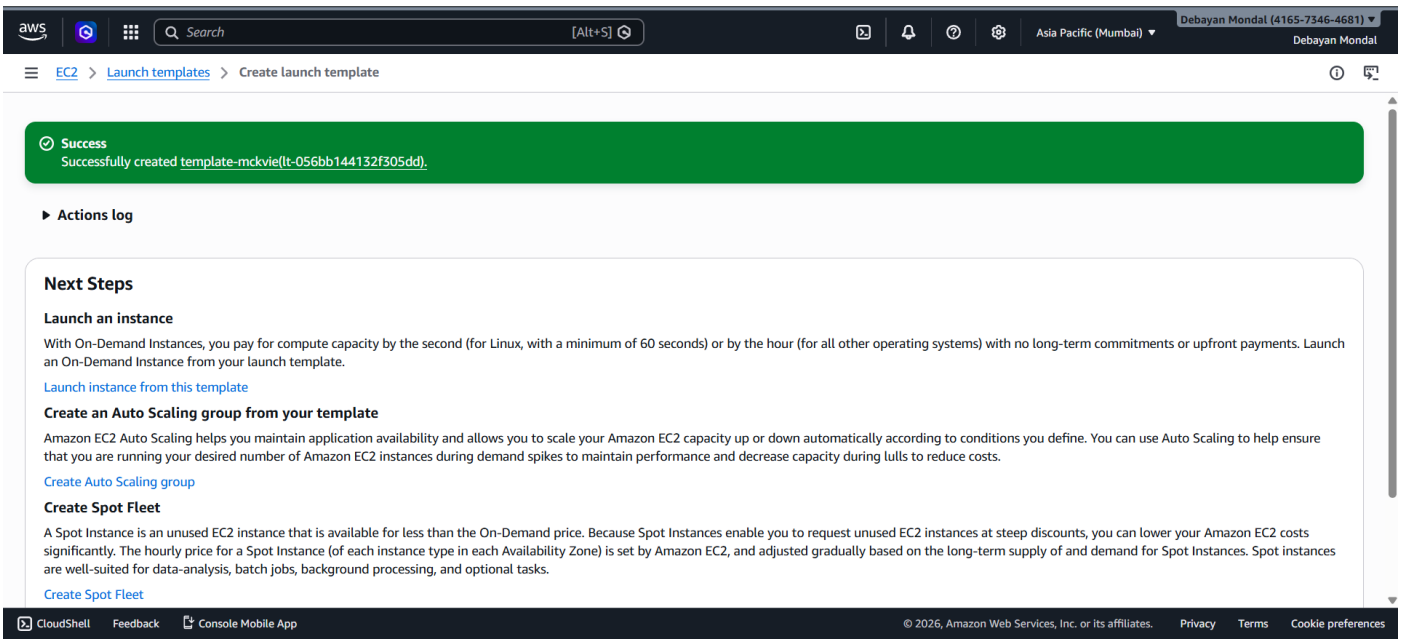
Step 7: In the User Data enter the following scripts has shown and click on **Create launch template** button.



Step 8: The template will be successfully created.



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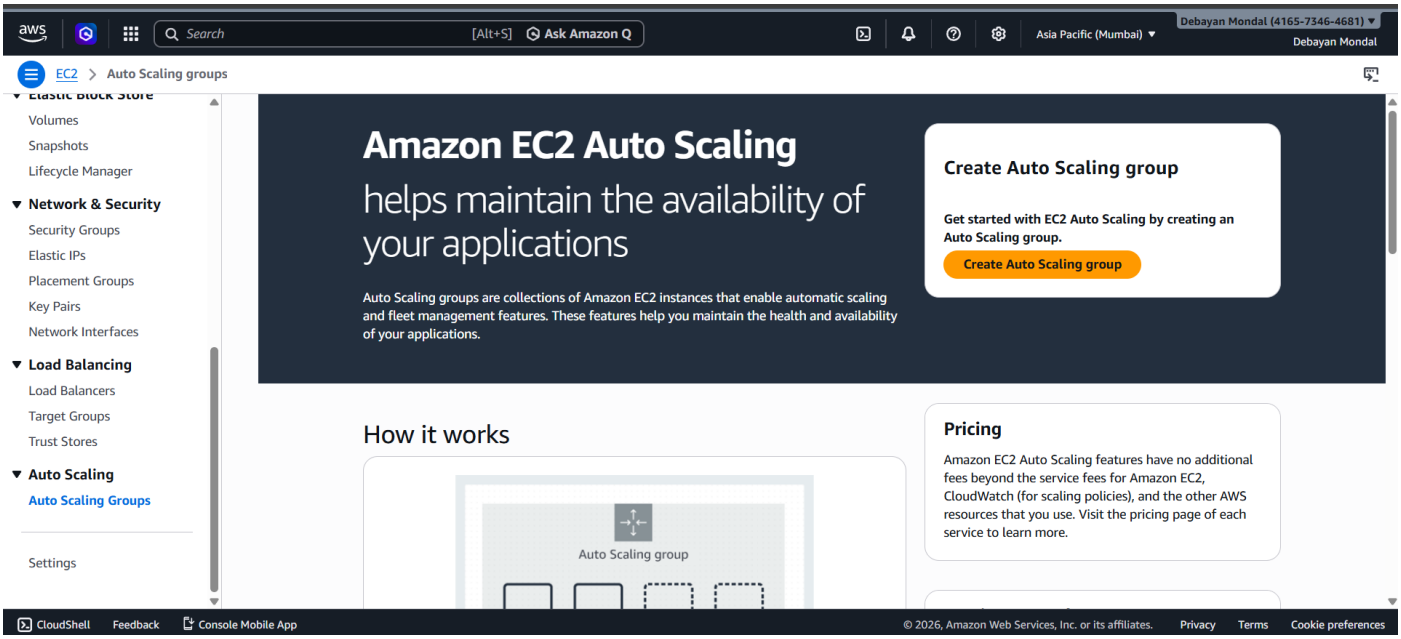
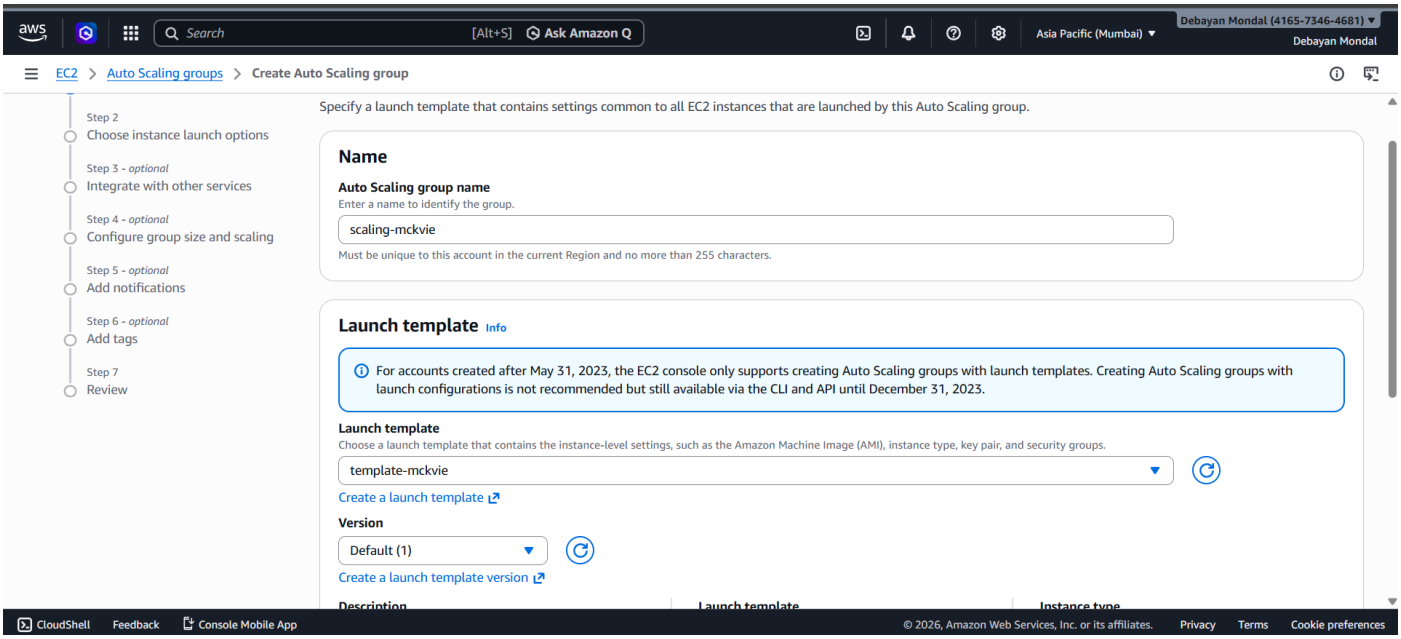
Step 9: From the side panel, click on **Auto Scaling Groups** then on that page click on the **Create Auto Scaling group** button.



Step 10: Enter the auto scaling group name and click **Next**.



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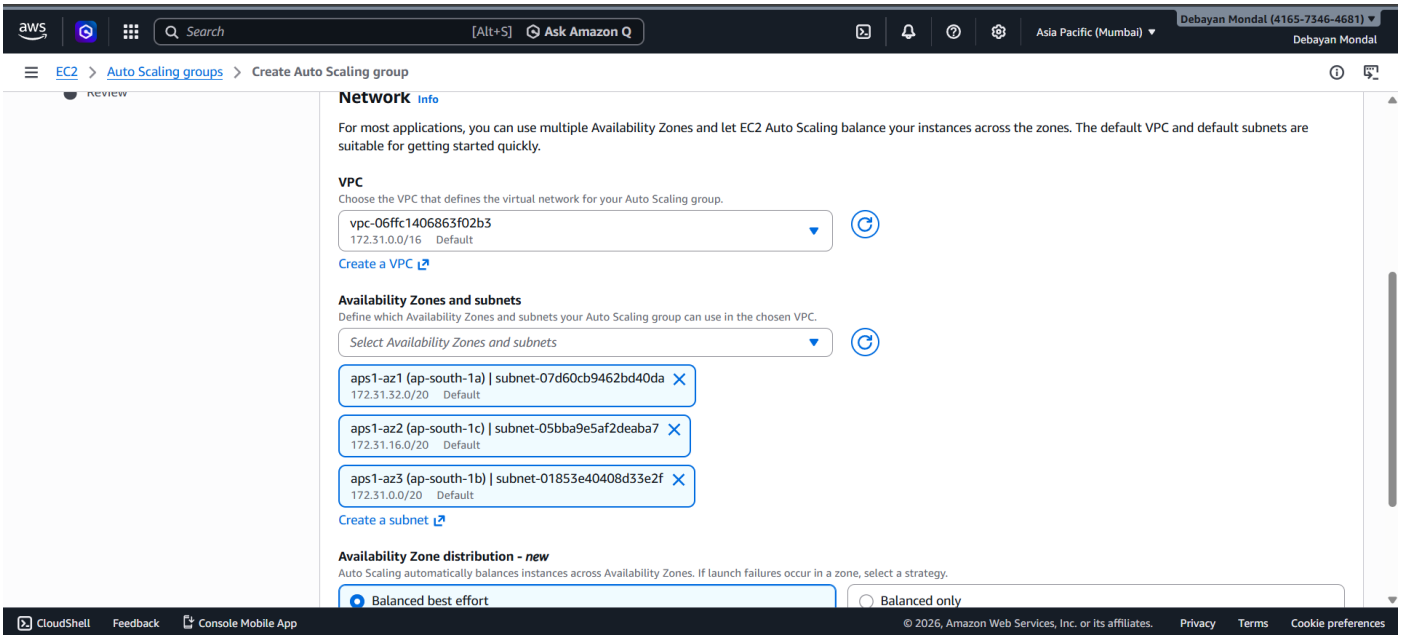
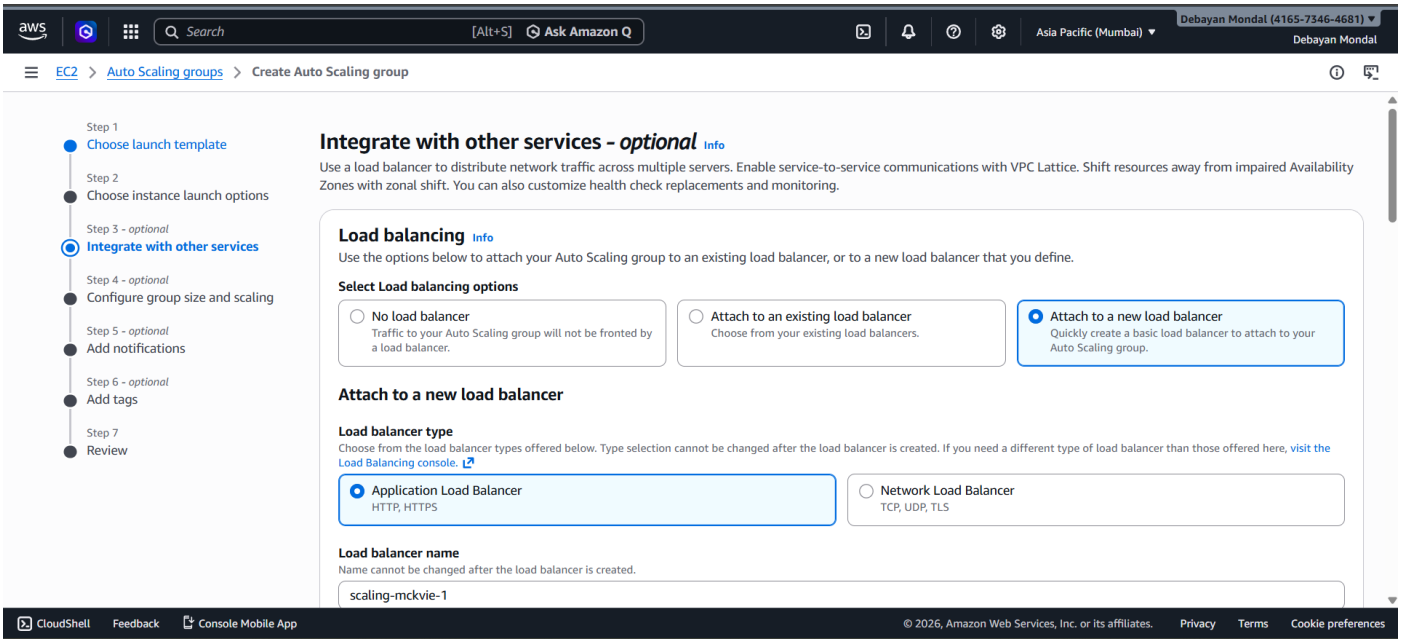
Step 11: In the next page, under the Availability zone and subnets select the 3 available zones and click on **Next**.



Step 12: In the next page, select **Attach to a new load balancer**.



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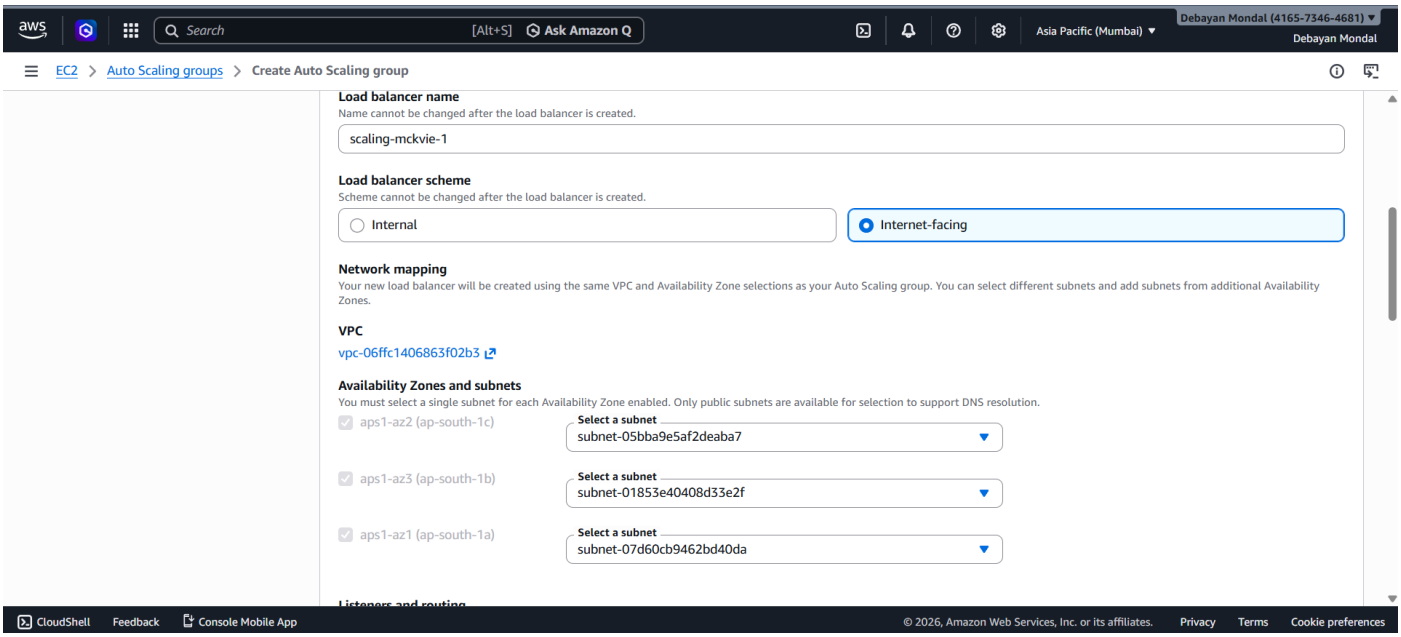
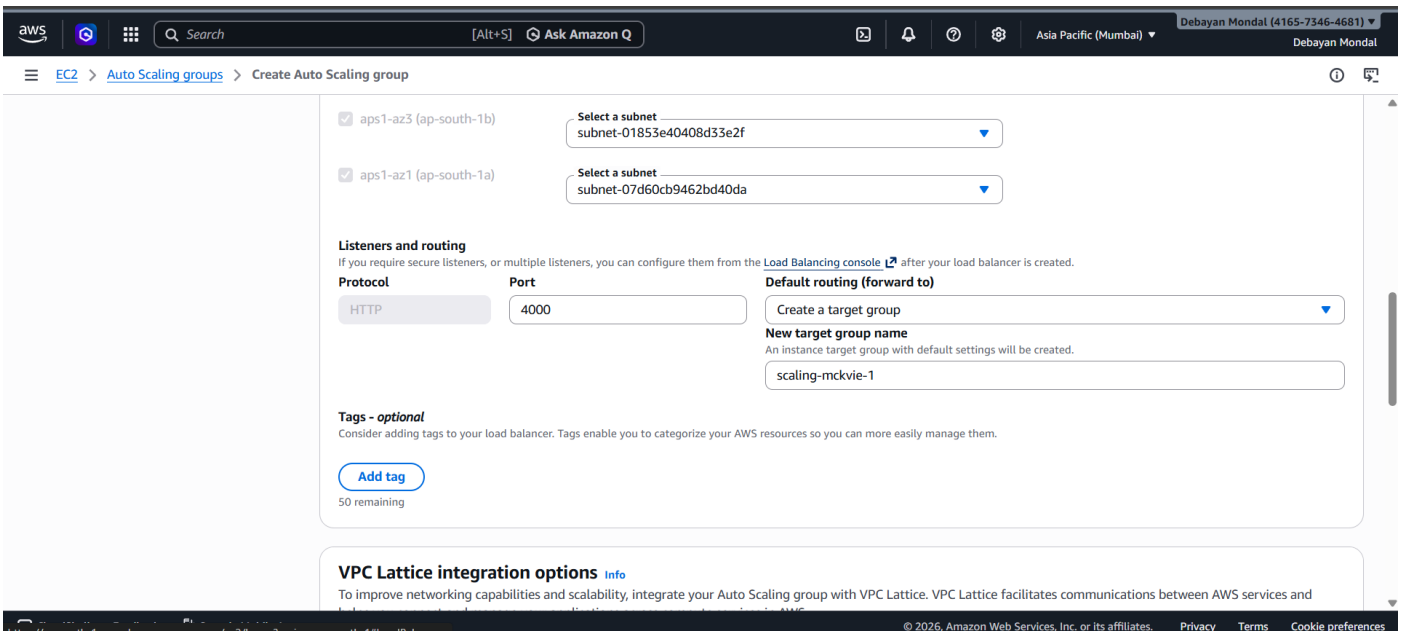
Step 13: Select Internet-facing.



Step 14: In the Default routing, click on **Create a target group**.



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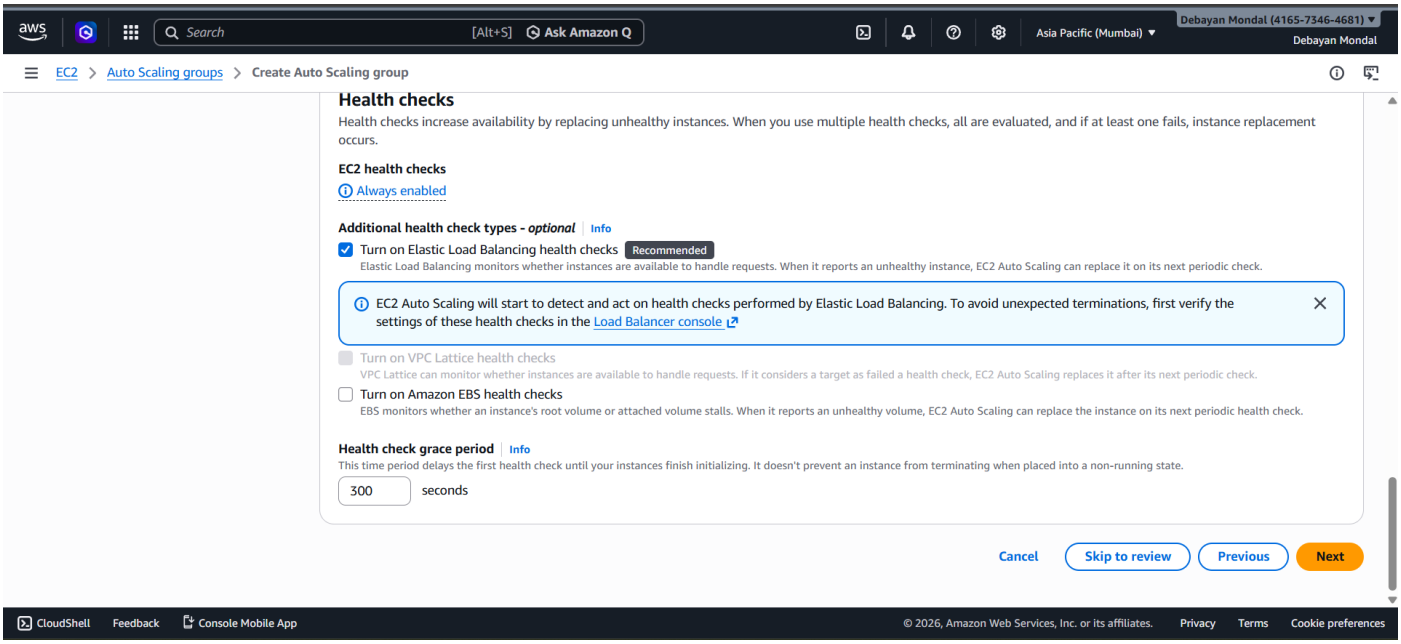
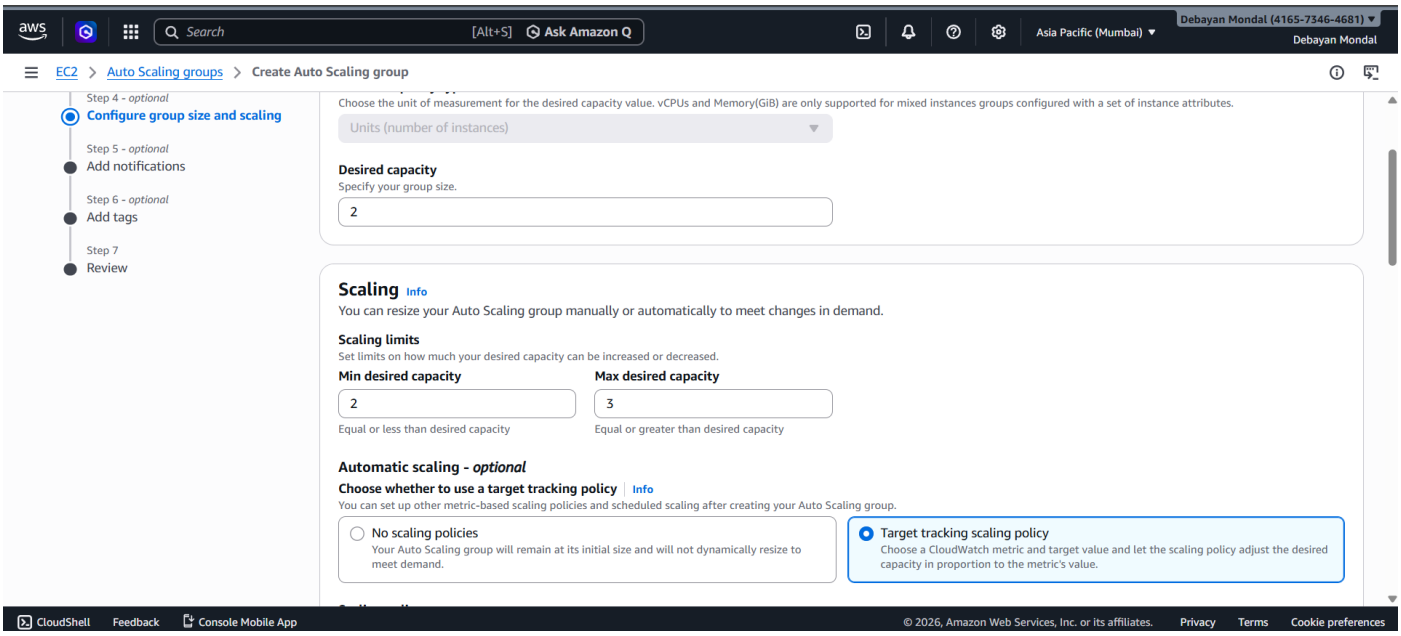
Step 15: Click on the checkbox **Turn on Elastic Load Balancing health checks**. Then click on the **Next** button.



Step 16: In the next page, increase the **Desired Capacity**, **Min desired capacity**, **Max desired capacity** by 1 and select **Target tracking scaling policy**.



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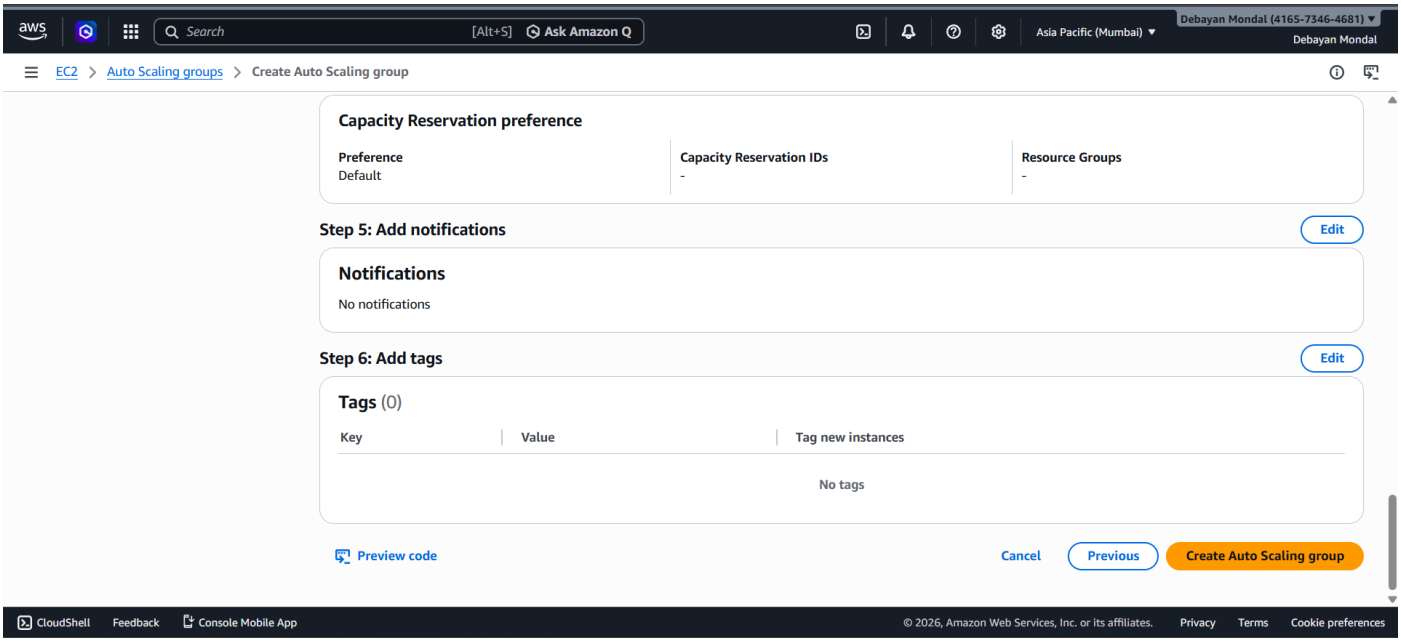
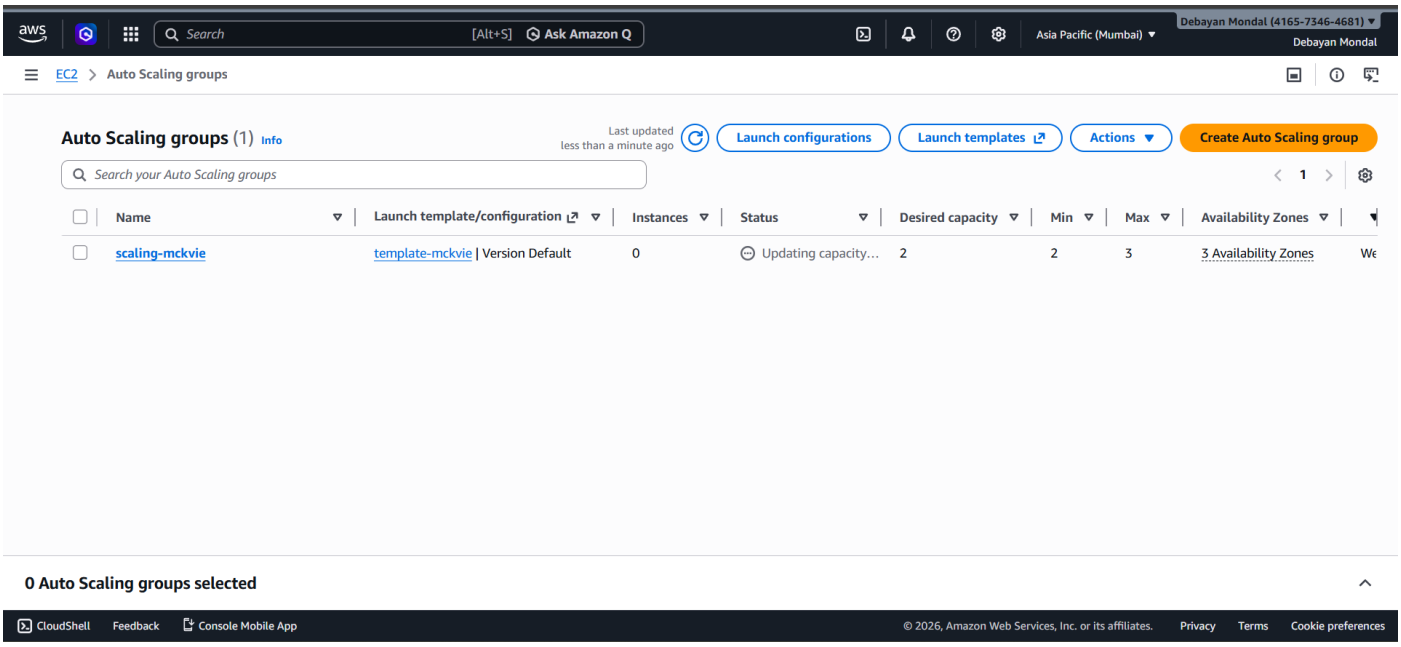
Step 17: Keep on clicking next till we reach the last Review page and then click on **Create Auto Scaling group** button.



Step 18: The Auto Scaling group will be created.



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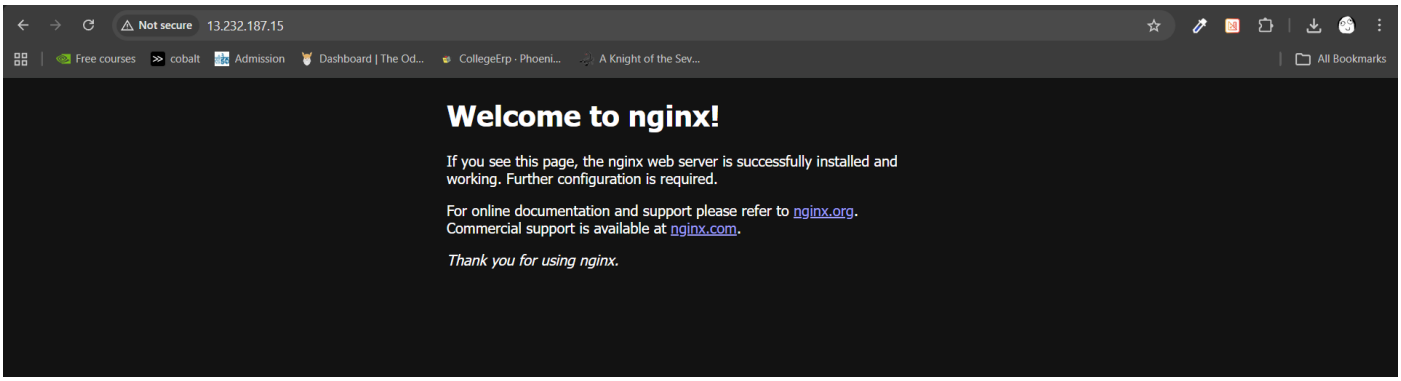
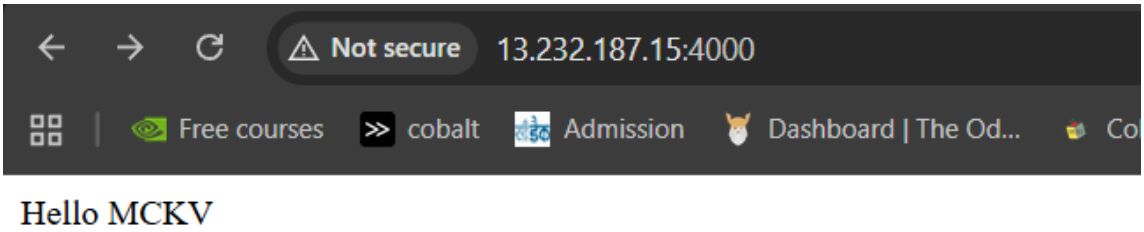
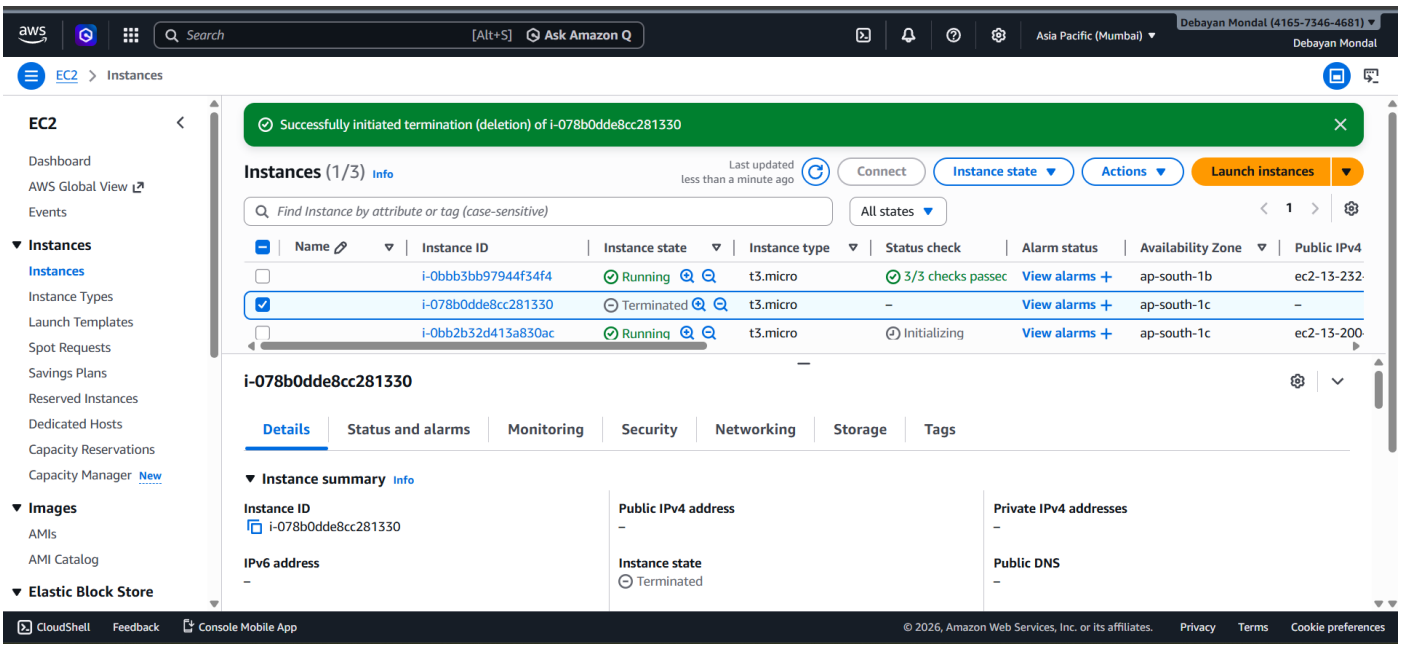
Step 19: Copy the IPv4 address of the instance created and paste it on a separate tab.



Step 20: Add :4000 to the url.



Step 21: If we delete on of the two Instances created, a newly instance will automatically be created.



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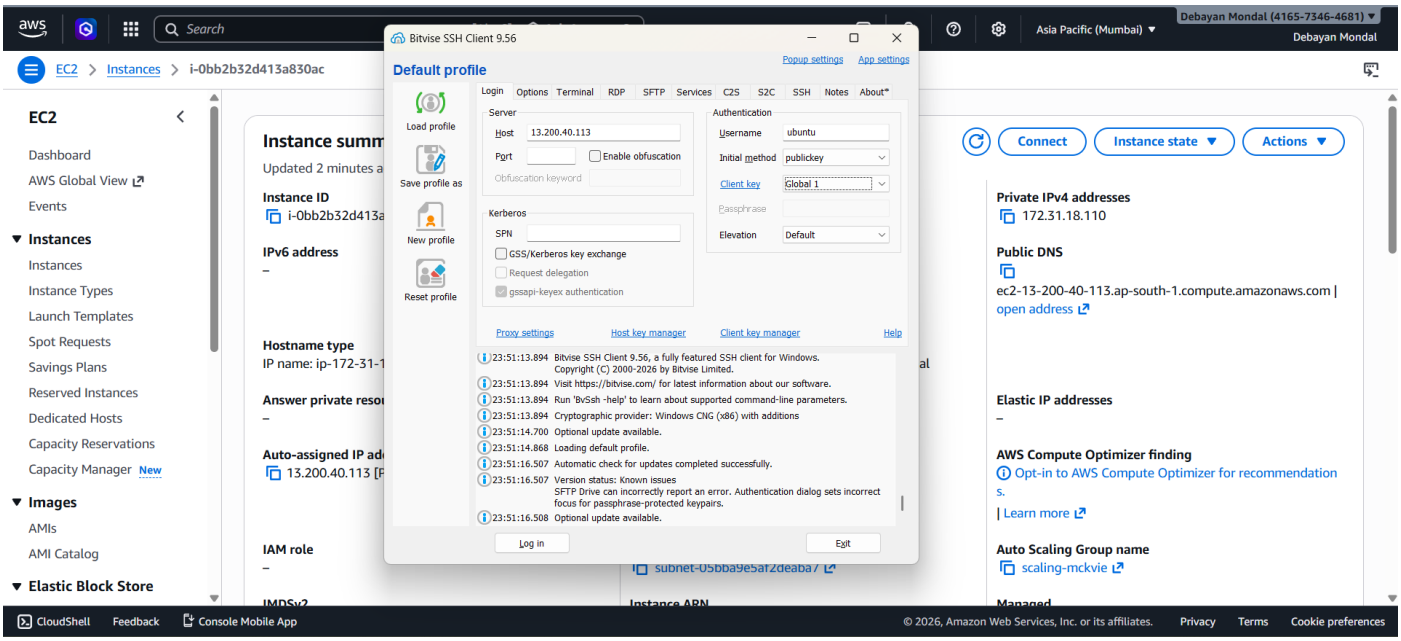
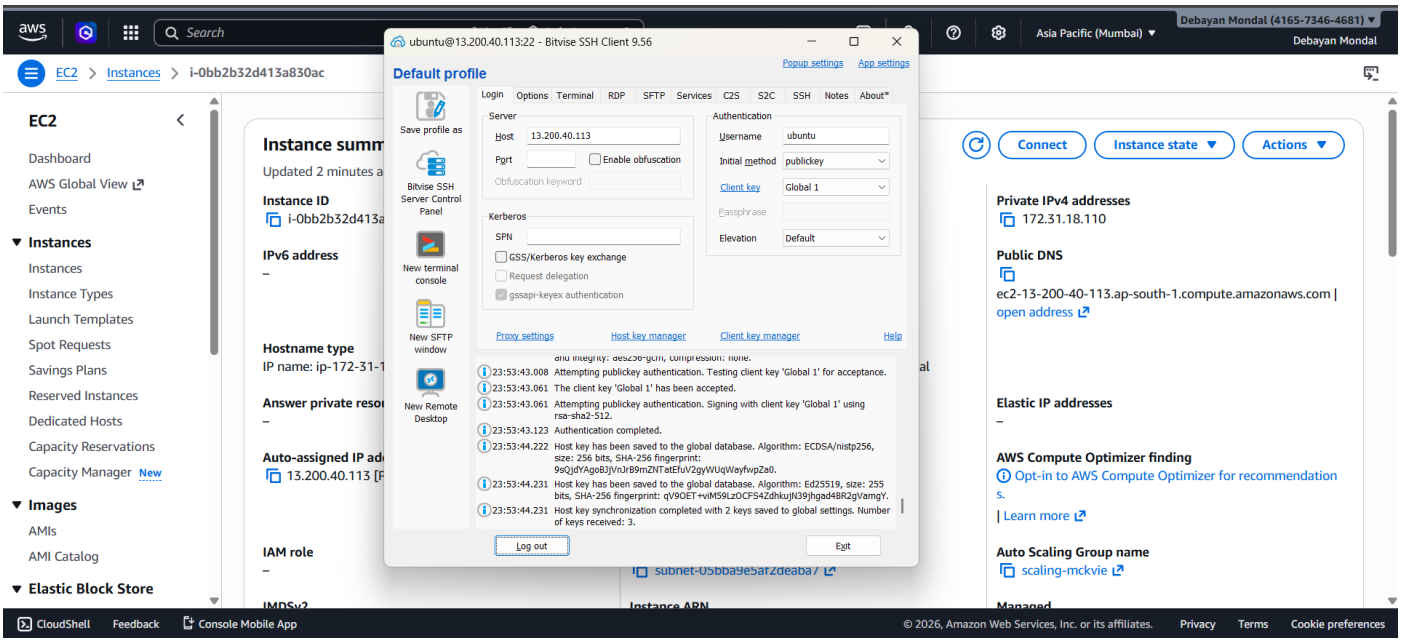
Step 22: Copy the IPv4 address of the Instance and Enter it into **Host** field of Bitvise. Also Enter the username as **ubuntu**, Initial method as **public key**, Client key as **Global 1** and click on Log in.



Step 23: It will be successfully connected with the instance. From the side panel, click on **New Terminal console.**



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Step 24: Check the working directory with command **pwd** and create a new shell file using the command **sudo nano debayan.sh**.

```
See "man sudo_root" for details.

ubuntu@ip-172-31-18-110:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-18-110:~$ sudo nano debayan.sh
```

Step 25: Write a script for an infinite loop inside the file.

```
ubuntu@13.200.40.113:22 - Bitvise xterm - ubuntu@ip-17
  GNU nano 7.2
#!/bin/bash
while(true)
do
        echo "Inside Loop"
done
```

Step 26: Give the necessary permission for the file to execute using the command **sudo chmod +x debayan.sh**.

```
debayan.sh
ubuntu@ip-172-31-18-110:~$ sudo chmod +x debayan.sh
ubuntu@ip-172-31-18-110:~$
```

Step 27: Execute the file using the command **sh debayan.sh**.

```
ubuntu@ip-172-31-18-110:~$ sh debayan.sh
```

Step 28: We will get an infinite loop.

```
ubuntu@13.200.4
Inside Loop
Inside Loop
Inside Loop
Inside Loop
Inside Loop
Inside Loop
Inside Loop
```

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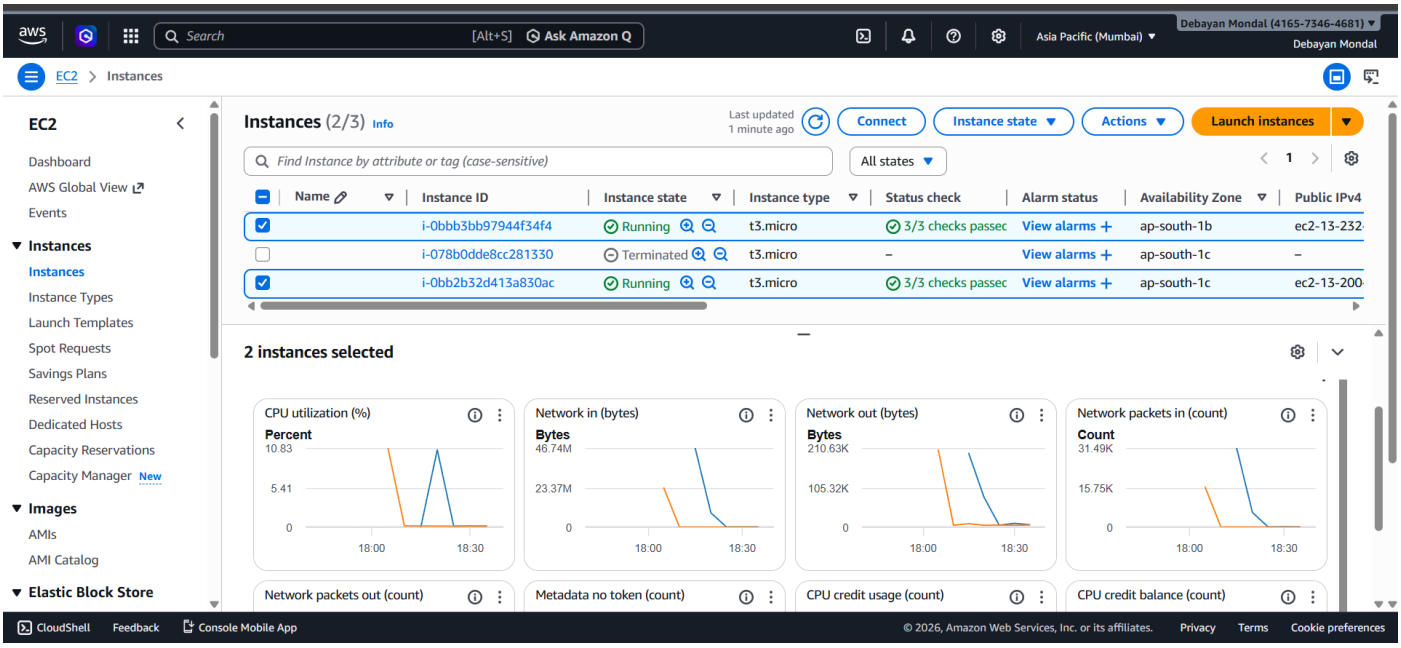
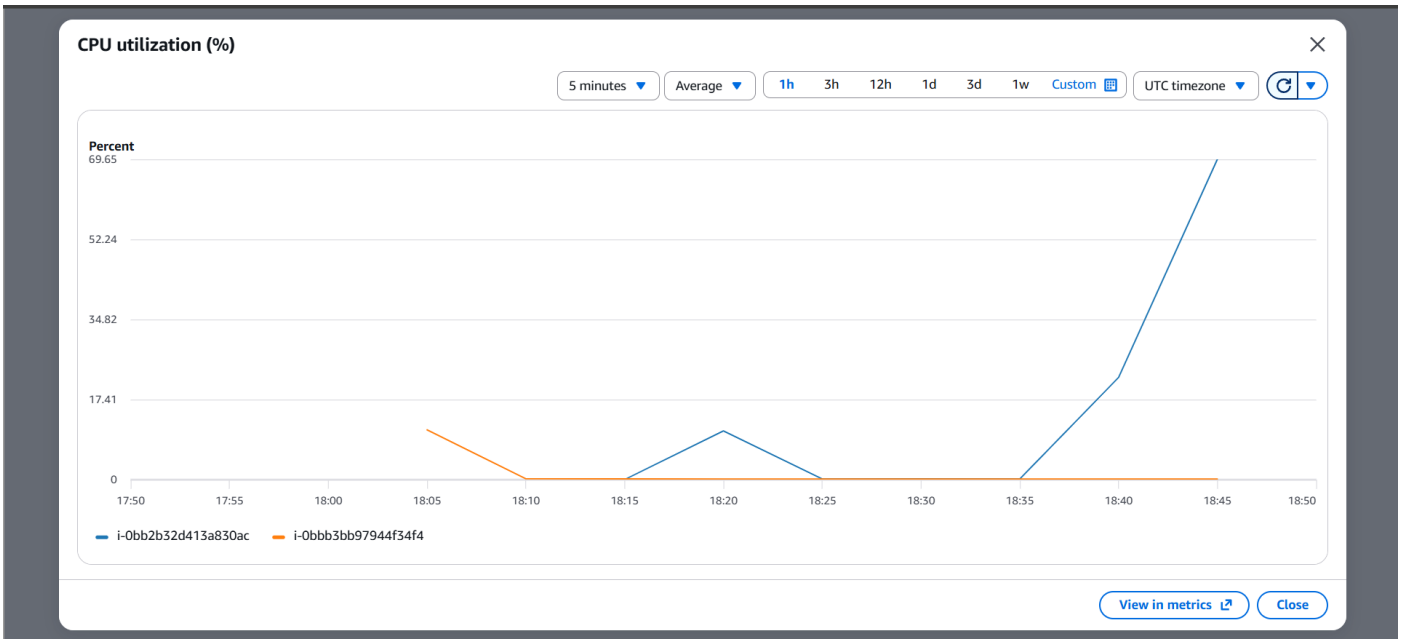
Step 29: Go to the instance page and select the 2 instances that has been created to observe the graphs between them.



Step 30: After a while the instance reaches and surpasses it's 50% CPU utilisation threshold. So, to take that load of the instance, the auto scaler spins up another instance to balance the load.